

DBSE FAST API CONNECTIONS

Step 1 — Upgrade pip tools

python -m pip install --upgrade pip setuptools wheel

```
PS C:\Users\akshitha Bushetty\Desktop\2nd year\Odd Sem\Database Magement System\Fast_API> python -m pip install --upgrade pip setuptools wheel
1
Requirement already satisfied: pip in C:\Users\akshitha Bushetty\AppData\Local\Programs\Python\Python313\Lib\site-packages (26.2.1)
Requirement already satisfied: setuptools in C:\Users\akshitha Bushetty\AppData\Local\Programs\Python\Python313\Lib\site-packages (84.0.0)
Requirement already satisfied: wheel in C:\Users\akshitha Bushetty\AppData\Local\Programs\Python\Python313\Lib\site-packages (0.48.0)
Requirement already satisfied: packaging>=24.0 in C:\Users\akshitha Bushetty\AppData\Local\Programs\Python\Python313\Lib\site-packages (from wheel) (26.3)
```

Step 2 — Install a compatible greenlet version

python -m pip install "greenlet==3.0.3"

```
PS C:\Users\akshitha Bushetty\Desktop\2nd year\Odd Sem\Database Magement System\Fast_API> python -m pip install "greenlet==3.0.3"
Collecting greenlet==3.0.3
  Using cached greenlet-3.0.3.tar.gz (182 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: greenlet
  Building wheel for greenlet (pyproject.toml) ... error
error: subprocess-exited-with-error

x Building wheel for greenlet (pyproject.toml) did not run successfully.
  exit code: 1
  > [119 lines of output]
    C:\Users\akshitha Bushetty\AppData\Local\Temp\pip-build-env-h03wku8r\overlay\Lib\site-packages\setuptools\dist.py:765: SetuptoolsWarning: License classifiers are deprecated.
      !!

      *****
      Please consider removing the following classifiers in favor of a SPDX license expression:

      License :: OSI Approved :: MIT License

      See https://packaging.python.org/en/latest/guides/writing-pyproject-toml/#license for details.
      *****

      !!
      self._finalize_license_expression()
      running bdist_wheel
      running build
      running build_py
      creating build\lib.win-amd64-cpython-313\greenlet
      copying src\greenlet\__init__.py -> build\lib.win-amd64-cpython-313\greenlet
      creating build\lib.win-amd64-cpython-313\greenlet\platform
```

```
    C:\Users\akshitha Bushetty\AppData\Local\Temp\pip-build-env-h03wku8r\overlay\Lib\site-packages\setuptools\dist.py
SetuptoolsWarning: License classifiers are deprecated.
    !!

    *****
    Please consider removing the following classifiers in favor of a SPDX license expression:

    License :: OSI Approved :: MIT License

    See https://packaging.python.org/en/latest/guides/writing-pyproject-toml/#license for details.
    *****

    !!
    self._finalize_license_expression()
    running bdist_wheel
    running build
    running build_py
    creating build\lib.win-amd64-cpython-313\greenlet
    copying src\greenlet\__init__.py -> build\lib.win-amd64-cpython-313\greenlet
    creating build\lib.win-amd64-cpython-313\greenlet\platform
    copying src\greenlet\platform\__init__.py -> build\lib.win-amd64-cpython-313\greenlet\platform
    creating build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\fail_clearing_run_switches.py -> build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\fail_cpp_exception.py -> build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\fail_initialstub_already_started.py -> build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\fail_slp_switch.py -> build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\fail_switch_three_greenlets.py -> build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\fail_switch_three_greenlets2.py -> build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\fail_switch_two_greenlets.py -> build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\leakcheck.py -> build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\test_contextvars.py -> build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\test_cpp.py -> build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\test_extension_interface.py -> build\lib.win-amd64-cpython-313\greenlet\tests
    copying src\greenlet\tests\test_gc.py -> build\lib.win-amd64-cpython-313\greenlet\tests
```

```

copying src\greenlet\platform\switch_ppc64_linux.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_ppc_aix.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_ppc_linux.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_ppc_macosx.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_ppc_unix.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_riscv_unix.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_s390_unix.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_sparc_sun_gcc.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_x32_unix.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_x64_masm.asm -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_x64_masm.obj -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_x64_msvc.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_x86_msvc.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\platform\switch_x86_unix.h -> build\lib.win-amd64-cpython-313\greenlet\platform
copying src\greenlet\tests\_test_extension.c -> build\lib.win-amd64-cpython-313\greenlet\tests
copying src\greenlet\tests\_test_extension_cpp.cpp -> build\lib.win-amd64-cpython-313\greenlet\tests
running build_ext
building 'greenlet.greenlet' extension
error: Microsoft Visual C++ 14.0 or greater is required. Get it with "Microsoft C++ Build Tools": https://visualstudio.microsoft.com/build-tools/
[end of output]

```

Step 3 — Install SQLAlchemy

python -m pip install sqlalchemy

```

> greenlet
PS C:\Users\akshitha Bushetty\Desktop\2nd year\Odd Sem\Database Magement System\Fast_API> python -m pip install sqlalchemy
Requirement already satisfied: sqlalchemy in C:\Users\akshitha Bushetty\AppData\Local\Programs\Python\Python313\Lib\site-packages (2.0.52)
Requirement already satisfied: greenlet>=1 in C:\Users\akshitha Bushetty\AppData\Local\Programs\Python\Python313\Lib\site-packages (from sqlalchemy) (3.5.5)
Requirement already satisfied: typing-extensions>=4.6.0 in C:\Users\akshitha Bushetty\AppData\Local\Programs\Python\Python313\Lib\site-packages (from sqlalchemy) (4.16.0)
PS C:\Users\akshitha Bushetty\Desktop\2nd year\Odd Sem\Database Magement System\Fast_API>

```

Step 4 — Install PyMySQL

python -m pip install pymysql

```

PS C:\Users\akshitha Bushetty\Desktop\2nd year\Odd Sem\Database Magement System\Fast_API> python -m pip install pymysql
Requirement already satisfied: pymysql in C:\Users\akshitha Bushetty\AppData\Local\Programs\Python\Python313\Lib\site-packages (1.2.0)
PS C:\Users\akshitha Bushetty\Desktop\2nd year\Odd Sem\Database Magement System\Fast_API>

```

Step 5 — Check everything

python -c "import fastapi, uvicorn, sqlalchemy, pymysql, greenlet; print('All installed successfully')"

```

PS C:\Users\akshitha Bushetty\Desktop\2nd year\Odd Sem\Database Magement System\Fast_API> python -c "import fastapi, uvicorn, sqlalchemy, pymysql, greenlet; print('All installed successfully')"
All installed successfully
PS C:\Users\akshitha Bushetty\Desktop\2nd year\Odd Sem\Database Magement System\Fast_API>

```

Step 6-Create a database in MySQL

CREATE DATABASE university;

USE university;

CREATE TABLE students (

id INT PRIMARY KEY,

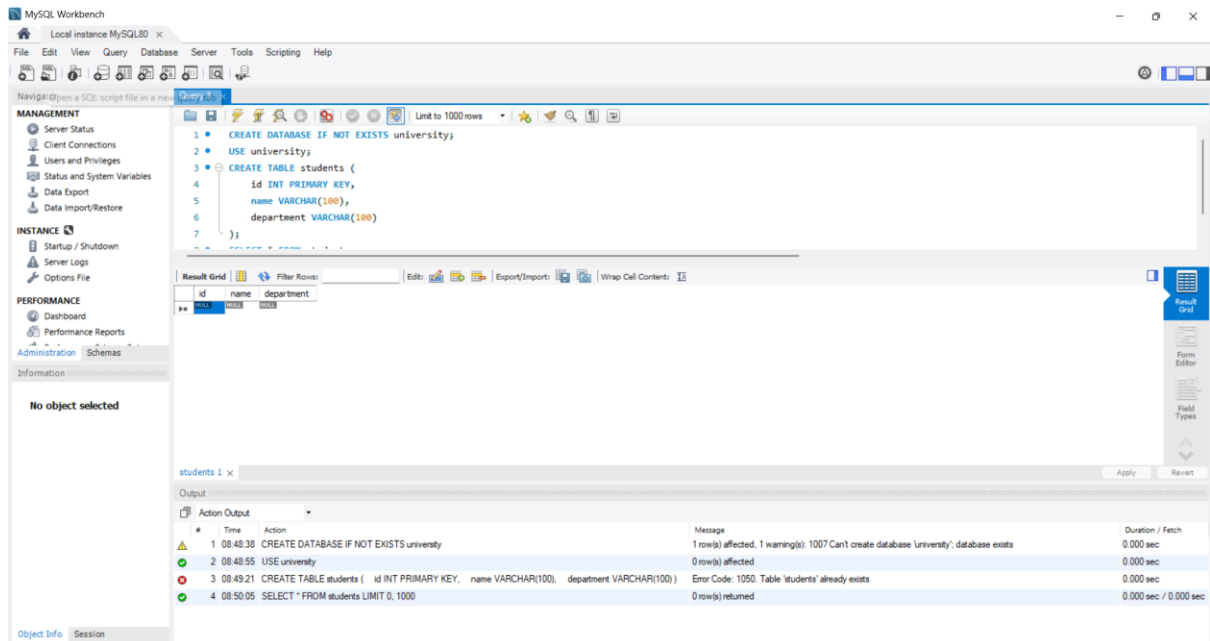
name VARCHAR(100),

department VARCHAR(100)

);

Check it:

SELECT * FROM students;



Step 7 Now create a GET API

from fastapi import FastAPI

from sqlalchemy import create_engine, text

app = FastAPI()

DATABASE_URL = "mysql+pymysql://root:123456789@localhost/university"

engine = create_engine(DATABASE_URL)

@app.get("/students")

def get_students():

with engine.connect() as connection:

result = connection.execute(

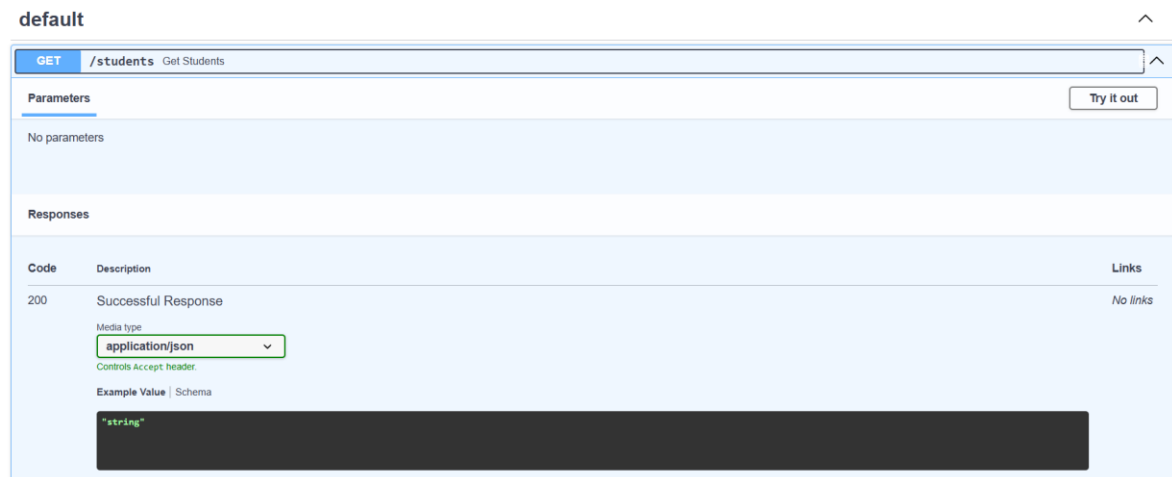
text("SELECT * FROM students")

)

```
students = result.mappings().all()
```

```
return students
```

FastAPI 0.1.0 OAS 3.1
/openapi.json



Step 8. Now POST can actually save data

```
from fastapi import FastAPI
```

```
from sqlalchemy import create_engine, text
```

```
from pydantic import BaseModel
```

```
app = FastAPI()
```

```
DATABASE_URL = "mysql+pymysql://root:your_password@localhost/university"
```

```
engine = create_engine(DATABASE_URL)
```

```
class Student(BaseModel):
```

```
    id: int
```

```
    name: str
```

```
    department: str
```

```
@app.get("/students")

def get_students():

    with engine.connect() as connection:

        result = connection.execute(

            text("SELECT * FROM students")

        )

        students = result.mappings().all()

        return students
```

```
@app.post("/students")

def create_student(student: Student):

    with engine.begin() as connection:

        connection.execute(

            text("""

                INSERT INTO students (id, name, department)

                VALUES (:id, :name, :department)

            """),

            {

                "id": student.id,

                "name": student.name,

                "department": student.department

            }

        )
```

```
return {  
    "message": "Student created",  
    "student": student  
}
```

FastAPI 0.1.0 OAS 3.1
/openapi.json

default

GET

/students Get Students

POST

/students Create Student

Schemas

HTTPValidationError > Expand all [object](#)

Student > Expand all [object](#)

ValidationError > Expand all [object](#)